

Artificial Intelligence and Blockchain Enabled Smart Healthcare System for Monitoring and Detection of COVID-19 in Biomedical Images

Imran Ahmed , Abdellah Chehri , and Gwanggil Jeon 

Abstract—Millions of individuals around the world have been impacted by the ongoing coronavirus outbreak, known as the COVID-19 pandemic. Blockchain, Artificial Intelligence (AI), and other cutting-edge digital and innovative technologies have all offered promising solutions in such situations. AI provides advanced and innovative techniques for classifying and detecting symptoms caused by the coronavirus. Additionally, Blockchain may be utilized in healthcare in a variety of ways thanks to its highly open, secure standards, which permit a significant drop in healthcare costs and opens up new ways for patients to access medical services. Likewise, these techniques and solutions facilitate medical experts in the early diagnosis of diseases and later in treatments and sustaining pharmaceutical manufacturing. Therefore, in this work, a smart blockchain and AI-enabled system is presented for the healthcare sector that helps to combat the coronavirus pandemic. To further incorporate Blockchain technology, a new deep learning-based architecture is designed to identify the virus in radiological images. As a result, the developed system may offer reliable data-gathering platforms and promising security solutions, guaranteeing the high quality of COVID-19 data analytics. We created a multi-layer sequential deep learning architecture using a benchmark data set. In order to make the suggested deep learning architecture for the analysis of radiological images more understandable and interpretable, we also implemented the Gradient-weighted Class Activation Mapping (Grad-CAM) based colour visualization approach to all of the tests. As a result, the architecture achieves a classification accuracy rate of 0.96, thus producing excellent results.

Index Terms—Artificial intelligence, blockchain, COVID-19, deep learning.

I. INTRODUCTION

ADVANCED digital technologies like the Internet of Things (IoT), 5 G telecommunication networks, and artificial intelligence that utilizes deep learning and Big Data analytics have emerged in the past decade. These technologies may have significant potential for various applications, particularly in health

care [1], [2]. For example, in the current pandemic situation, the deadly COVID-19 virus, which paralyzed access to international healthcare systems through unprecedented lockdowns and implemented social distancing, has fast revved the growth of these digital technologies to fulfil various healthcare demands globally [3], [4], [5], [6]. In addition, coordination of high-level operations such as rapid contact tracing, mass screening of large populations, supply chain control and management for drugs and vaccines, telemedicine consultations, and E-commerce development has driven a broad spectrum of applications using these advanced digital technologies [7]. Blockchain is considered likely to have the most significant impact on the world in the coming years. Its applications have been developed well beyond cryptocurrency and bitcoin, securely recording cryptocurrency transactions. It helps to make more fairness, transparency, precision, privacy, and security; this distributed ledger methodology has released numerous new applications in various sectors such as healthcare, telecommunications, transportation, agriculture, and automotive manufacturing. Moreover, it is the basis of advanced digital technology that combines multiple technologies to provide a smart and innovative solution for different real-life applications.

Blockchain could substitute conventional distributed database management systems in healthcare, commonly client-server databases with relational input or Structured Query Language [8]. Although traditional databases are regular platforms in healthcare systems, they have many constraints, for example, the vulnerability to external adversaries (hacking), the inability to keep peer-to-peer data sharing, and the lack of a narrow audit trail. With unique characteristics of Blockchain technology that involve traceability, transparency, and non-repudiation defined as the incapability to deny signature validity, disintermediation refers to reducing mediators from a decision-making strategy) and immutability. These problems are now manageable with substantial benefits over traditional platforms. By combining blockchain technology, decentralized healthcare database management systems can be constructed. These systems can coordinate on-chain circumstances, like recording transactions on the blockchain ledger and off-chain circumstances, for example, external events, that are usually too extensive to keep on the Blockchain.

Although overall implementation and adoption of Blockchain in healthcare are still limited by different technical, organizational, and government obstacles, however, the COVID-19

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pandemic has obviously highlighted the requirement for secure, decentralized, multi-purpose systems and platforms for coordinating the large-scale transmission of sensitive and essential data or information, including the transmission of diseases, monitoring of the vaccination status, contact tracing, and COVID-19 health certificate issuance. These demands deliver a powerful momentum for a concerted action to force more significant applications and overcome a few of these obstacles to the general adoption of Blockchain technology.

Moreover, all health warehouse servers, for example, the cloud used for the coronavirus data recording, collection, and sharing, would be automatically managed by the Blockchain networks. Information used over these networks is consistently recalled again on the Blockchain networks for monitoring [9]. More significantly, connected with intelligent communication technology reduces the dependence on main servers to assure the fairness of transactions among patients, hospitals, and administrations and gives effective and adequate access to data and information. Data collection is an essential stage of disease analysis; thus, Blockchain ensures collected data's reliability [10]. It can present several potential solutions for coronavirus or COVID-19 pandemic-related services, including safe day-to-day operations, outbreak monitoring, disease controlling, pharmaceutical supply chain, and donation tracking.

Blockchain can assist in combating pandemics situations by allowing earlier monitoring and detection of spells, securing the order of collected medical data, and providing a dedicated medical and pharmaceutical supply chain throughout the outbreak monitoring. Furthermore, AI presented intelligent and innovative systems and solutions for classifying and detecting symptoms caused by the coronavirus. AI can give support for pandemic fighting through five significant applications, i.e., outbreak analysis, vaccine/drug development, coronavirus prediction and detection of any future coronavirus or COVID-19-like outbreaks. These systems and solutions facilitate medical experts in the early diagnosis of diseases and later in treatments and sustaining pharmaceutical manufacturing.

In this work, a smart blockchain and AI-enabled system are presented for the healthcare sector that helps to combat the coronavirus pandemic [11], [12]. Furthermore, a new multi-layer sequential deep learning-based architecture is developed for the detection of the virus in Chest X-ray images [13], which is further integrated with Blockchain [14]. Similarly, it provides promising security solutions and reliability of data collection platforms to deliver high-quality COVID-19 data analytics results [15]. Data augmentation is applied to improve the results and to provide more explainable and interpretable results. Grad-CAM approach is also used. This work also reveals the characteristic of Blockchain networks that can help to stop un-secure data sharing between stakeholders, including patients and health entities such as doctors, government organizations, and healthcare providers. The primary contribution of the work is presented as follows:

- To develop an intelligent and smart AI and Blockchain-enabled system for monitoring the COVID-19 virus in medical images.
- To propose a multilayer sequential deep learning model is proposed for the classification of Chest X-ray images.

- To apply data augmentation techniques to improve the classification results and overcome the overfitting problem.
- To provide a more explainable analysis of the diseases in X-ray images and classification results, the Grad-CAM algorithm is applied.
- To make a comparison of the developed method with other algorithms in terms of accuracy.

The rest of the paper is organized as follows: Section II delivers a summary of related work. Then, in Section III, a smart AI and blockchain-enabled system developed for the monitoring of COVID-19 is presented. Next, the details of the data set and experimental outcomes are explained in Section IV. The conclusion and future guidelines of the work are finally provided in Section V.

II. RELATED WORK

Advanced digital technologies such as IoT, AI, 5 G communication [16], [17], [18], [19], and [20] along with Big Data analytics [15], and IoT [21], [22], [23], and [24] helps to understand variables, behaviours, and data trends utilizing a broad range of algorithms and techniques that are substantially better than human performance. Initial investigations studied different techniques like statistical methods, data mining, and artificial intelligence-based approaches (machine learning), supported segmentation methods (parallel, classified processing, statistical rationalizing), and distribution of Big Data applications for various types of control studies like prediction and risk [25]. This section presented a review of various artificial intelligence (machine learning based algorithms) studied by researchers for the analysis and detection of disease. In [26], authors provided a comparative analysis of soft computing and machine learning paradigms to estimate the outbreak. Ahmed et al. [15], designed a framework applying Big Data analytics and IoT for the analysis, investigation, and prediction of COVID-19.

Punn et al. [27] employed machine learning and deep learning to identify growth and divine the future measure of the pandemic. Authors performed various types of analysis using different pandemic symptoms. In [25], researchers analyzed clinical data (textual) into four separate categories by using conventional and ensemble machine learning methods. Authors in [28] presented a broad survey on Big Data tools and machine learning algorithms utilized in various healthcare practices.

Authors in [29] applied machine learning to forecast and evaluate the pandemic. Behnam et al. [30], presented a data analytics system for predicting the coronavirus. Authors compiled a data set that mainly contains the number of recovered, confirmed, and death cases reported every day using machine learning algorithms. Khakharia et al. [31], produced a prediction method for the COVID-19 outbreak; the system is developed for the top ten extremely dense populated regions. Applying various machine learning techniques, the models predict the number of newly reported cases expected to rise for five consecutive days.

Jiang et al. [32] added a machine-learning method that predicts COVID-19 infected patients. The system produced 80% accuracy on sample data of 53 patients used for training. Rao et al. [33] produced a method based on artificial intelligence for identifying subjects with COVID-19 using a mobile phone.

Finally, authors in [34] presented a modeling method to assess high-risk patients in the early stage using machine learning with three clinical features. Different researchers have done various research work on the prediction of COVID-19 [35]. The study mainly focused on a conceptual method for different applications, using data mining methods [35].

Li et al. [36] developed a regression model for the prediction of the pandemic's spread. Pinter et al. [37] added a hybrid model for outbreak prediction in Hungary. The model made no supposition on the transmission of the infection and pandemic; rather, it divides the time series analysis of the infected and fatality cases. Ahmed et al. [38], offered a comprehensive study based on machine learning techniques used for the investigation of the pandemic. The authors analyzed the difficulties and presented suggestions to the experts that might improve methods used to predict positive cases.

Rustam et al. [39], described the potential of machine learning techniques to find the number of predicted COVID-19 patients. In particular, four traditional prediction models have been practiced. [40], performed research identifying the pandemic impacts worldwide utilizing machine learning algorithms. The authors suggested an additive regression design that specialists typically evaluate with field knowledge about the time series. Ahmed et al. [41] studies machine learning method for genome sequence analysis of the COVID-19 infection.

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Hedman et al. [42] applied and compared various neural network models for the classification of coronavirus in Chest X-ray images. [43] introduced a CNN based system for the analysis and classification of chest X-rays into coronavirus, pneumonia, and normal images. [44] studied residual neural network model for the examination of X-rays. Pathak et al. [45] made a deep learning based technique along with transfer learning for the classification of CT scan images into infected cases. Ultimately, [46] introduced convolutional neural networks based multi-objective differential evolving model to distinguish the coronavirus cases utilizing CT scan data of chest. [47], proposed a healthcare system based on artificial intelligence for the detection of coronavirus using chest CT scans or X-ray images.

Muhammad et al. [48], presented a multi-layers fusion-based model for COVID-19 and non-COVID-19 classification using ultrasound images of lungs. [49] presented a neural network-based system using contrastive loss for the detection of coronavirus in X-ray images. [50] introduced a detection system based on transfer learning and neural networks for the analysis of chest X-ray images. [51] made COVID-19 identification using their own developed data set, including 1,144 X-rays, of coronavirus, pneumonia, and normal images.

Ahmed et al. [52], used a Faster-RCNN model for monitoring of COVID-19 virus in X-ray images. [53] and [54] trained 2D CNN model for data set collected from [55]. In [53], authors combined different pre-trained designs with regularization of support vector machine, while in [54], authors introduced a network by leveraging the power of capsule networks with different architectures to increase classification accuracies. [56] produced a deep learning-based diagnosis framework to assist medical experts in identifying patients having symptoms of COVID-19 virus and pneumonia in CT scan data.

It is concluded, from the above discussion, that various artificial intelligence-based methods [57], [58], [59], [60], [61], and [62] has been used by researchers that helped in analyzing, detecting, and predicting the coronavirus disease. However, it still needs many research efforts in order to enhance or improve its performance. Therefore, in this work, we developed a deep learning-based method for the classification of COVID-19 Chest X-ray images. The developed system takes advantage of blockchain technology and gives an intelligent and secure system for data analysis.

III. METHOD

This paper presents a smart healthcare system that combines Blockchain and AI for coronavirus monitoring. The operational flowchart of the developed architecture is outlined in Fig. 1. The architecture is categorized into four main layers: the data collection, blockchain functional, AI module, and stakeholders layers. The workflow is illustrated as follows:

First, the data is collected in medical images (Chest X-rays) from various clinical labs and hospitals. The collected data is then needed to be secured for protection by utilizing the blockchain layer. This layer provides a secure network for data analytics and artificial intelligence module. By using reliable prediction and accurate analysis capacity on the collected data, the AI module helps detect and classify coronavirus in radiology images. Finally, the main objective of the architecture is to provide a secure medium to stakeholders; thus, the top order layer contains parties that mainly include patients, doctors, government organizations, hospitals, and healthcare stakeholders who are assisting from blockchain and AI-enabled systems. The data collected from hospitals is communicated with the data center via blockchain networks. AI tools are employed in the data center, and deep learning architecture is employed to classify Chest X-ray images. Finally, blockchain is applied to set a secure connection to exchange the analyzed results between stakeholders via the decentralized ledger. The details of the proposed AI model based on data pre-processing, data augmentation, deep learning architecture, and the Grad-CAM method that is applied for the classification of chest X-rays are provided as follows:

A. Data Pre-Processing and Data Augmentation

First, the data pre-processing is performed at the input layer. The pre-processing method typically takes the input image as a vector. The input image is also normalized to enhance the developed convolutional neural network rate during the training.

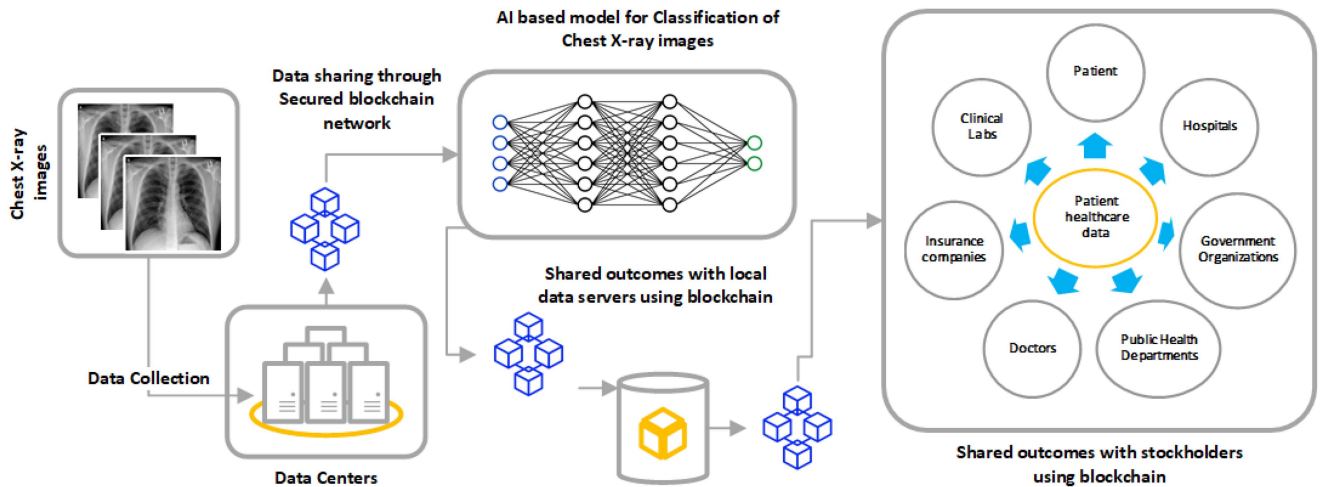


Fig. 1. Proposed innovative healthcare system illustrative based on blockchain and AI used for monitoring and classification of COVID-19 in medical images. The first layer collects data using securing blockchain network; the data is then forwarded to AI based model for classification. The outcomes or results are delivered to the local data servers and then provided to different stakeholders using blockchain technology.

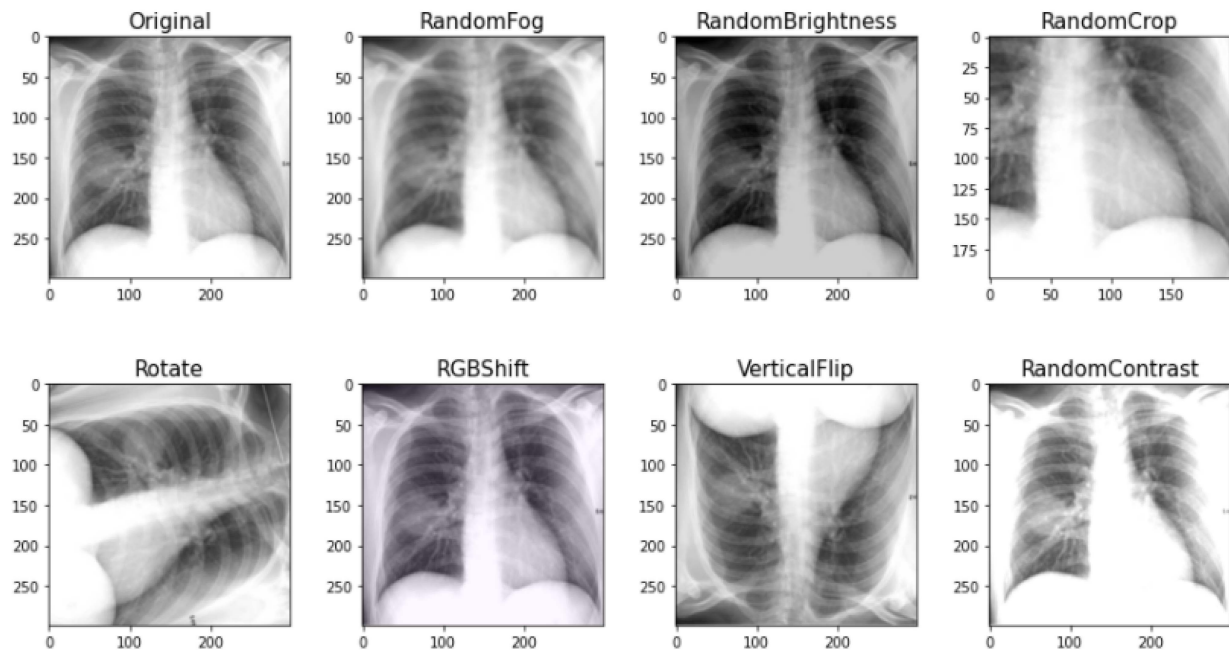


Fig. 2. Results of data augmentation on some example images of the data set. Different geometric, colour, and contrast transformations can be seen in COVID-19 X-ray images.

In order to improve the training results and enhance the accuracy of the deep learning architecture, data augmentation is applied. Various techniques, including “Random Fog”, “Random Brightness”, “Random Crop”, “Rotation”, “RGB Shift”, “Vertical Flip”, and “Random Contrast” are used to improve the amount of data. The results of the data augmentation methods are visualized in Fig. 2. From sample images, significant transformation can be seen in images.

B. Proposed Deep Learning Based Model

The proposed deep learning model is based on Convolutional Neural Network (CNN) architecture, consisting of multiple layers, including input, convolutional, pooling, fully connected,

and an output layer. As described in Fig. 3, the top layer of a typical CNN architecture is the input layer, followed by a pile of convolutional layers, batch normalization, and max-pooling layers or configurations that combine numerous convolutional and max-pooling layers. The final layer is the fully connected layer that is used for classification. The SoftMax classifier is used to determine the virus in chest X-ray images. There are two phases in convolutional neural network architecture: feature extraction or feature learning and classification. The first phase comprises convolutional, max-pooling, batch normalization, and fully connected layers. The input layer pre-processes the input images while feeding the raw data into the convolutional neural network. The convolutional layers are intended to produce feature maps using a convolution operation based on a collection

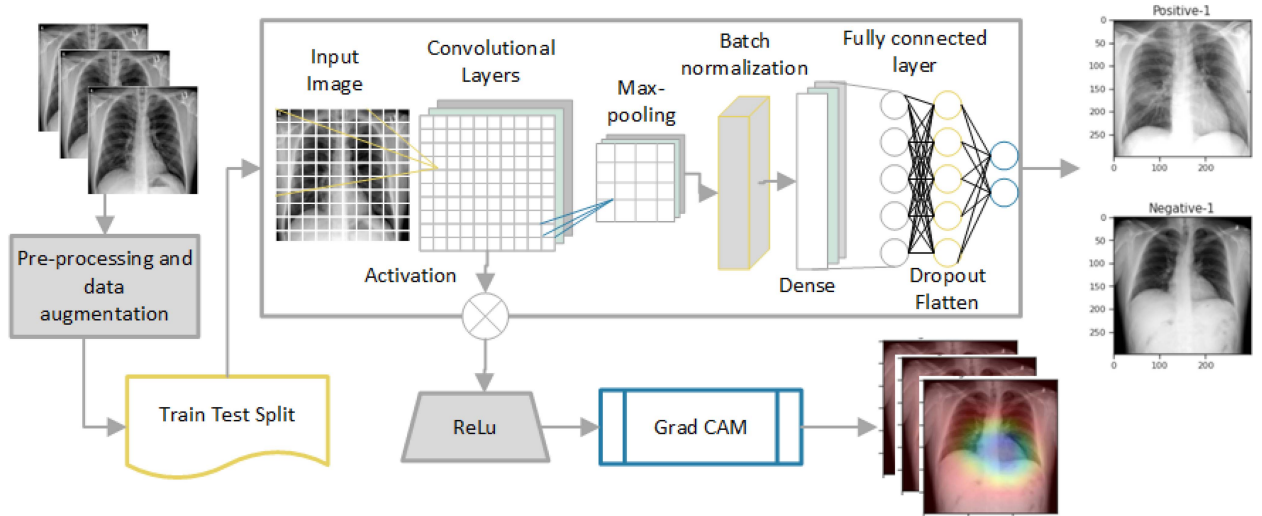


Fig. 3. A proposed deep learning model was used to classify Chest X-rays, and a simple explanatory diagram of the Grad-CAM method was used to explain the results to determine the parts of an input image that most impact the classification rate.

of weighted filters. The ReLU activation function is used to construct feature maps of the output from convolutional layers. The ReLU function is defined as;

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases} \quad (1)$$

In (1), x represents the activation function (input). The sub-sampling process is applied to filter the most important by using the max-pooling layers of the perception field and extracting the most representative characteristics in the region of interest. It reduces the output feature map or scale, thus lowering the number of parameters needed by the architecture and helping to maintain translation in variance. The final layer of the CNN is the fully connected layer that aims to translate or decode the last features extracted. The convolutional neural network architecture also learns these features. The extracted feature maps are categorized by applying the SoftMax classifier. Considering that the measurements of the input M and output vectors N , The fully connected layer computed number of parameters are as follows:

$$Q = MN + N \quad (2)$$

The extended SoftMax regression method is used for classification and dealing with two-state polymorphic problems. The training set comprises a set of sample labels $(x^{(1)}, y^{(1)}), \dots, (x^{(k)}, y^{(k)})$. The label values y are equal to 0 or 1, and the input features are $x^{(i)} \in R_n + 1$. The logistic regression function is measured as follows;

$$h\theta(x) = \frac{1}{1 + e^{-\theta^T x}} \quad (3)$$

Where θ represents model parameters, and then the $J(\theta)$ that represents the minimum loss function behind the training is provided as;

$$J(\theta) = \frac{1}{k} \left[\sum_{i=1}^k y^{(i)} \log h\theta(x^{(i)}) + 1y^{(i)} \log 1 - h\theta(x^{(i)}) \right] \quad (4)$$

We consider n conditions and corresponding labels. The SoftMax classifier uses these output labels. The training set comprises of the $(x^{(1)}, y^{(1)}), \dots, (x^{(k)}, y^{(k)})$ label samples, where $y^{(i)} \in \{1, 2, \dots, n\}$. For provided training samples x with n classes, the probability occurrence of state i is equal to $p(y = i|x)$. The problem of overfitting often occurs in classification applications; thus, the regularization methods, i.e., dropout and batch normalization, are applied in this paper.

The dropout layer temporarily drops a few neurons at the output with a particular probability p . The learning and training values are later shown in the left neurons with a possibility of $1 - p$. All neurons are fixed, and when the subsequent training is completed, some neurons are chosen randomly with p probability to temporarily not selected neurons, and the method is replicated. Utilizing the dropout process is similar to selecting a diverse architecture configuration for separate training, decreasing the adaptability and dependency of neurons on each other and improving the architecture's robustness. Next, batch normalization is conducted to normalize the layer. The output characteristics are normalized into the data with mean 0 and mean-variance 1 for the prior layer and then for the following layer input of the architecture.

The proposed CNN model design is outlined in Fig. 3. We considered Chest X-ray images of size $224 \times 224 \times 3$ pixels for input to the model. Our proposed model consists of three Conv layers, two max-pooling, three dense layers, and dropout layers. After input to the model starting with a small number of filters, first conv1 is used, which has 128 filters of size 3×3 , and each Conv layer is followed by max-pooling layers of fixed size 2×2 , which downsample the input for the subsequent Conv layers. Next is the Conv2 of filter size $64 \times 3 \times 3$ followed by max-pooling and then Conv3 of filter size $64 \times 3 \times 3$ for a down sampling of a feature map using max-pooling. In every path, zero padding by 1 pixel is used in the Conv1 layer to hold a similar spatial dimensional element map. The input to the conv2 layer is $16 \times 2 \times 2$, and after the output is given $32 \times 3 \times 3$, which applies 32 filters of size 3×3 , the same outcome is input to other layers. The conv3 layer has 64 feature maps of dimension 3×3 that can

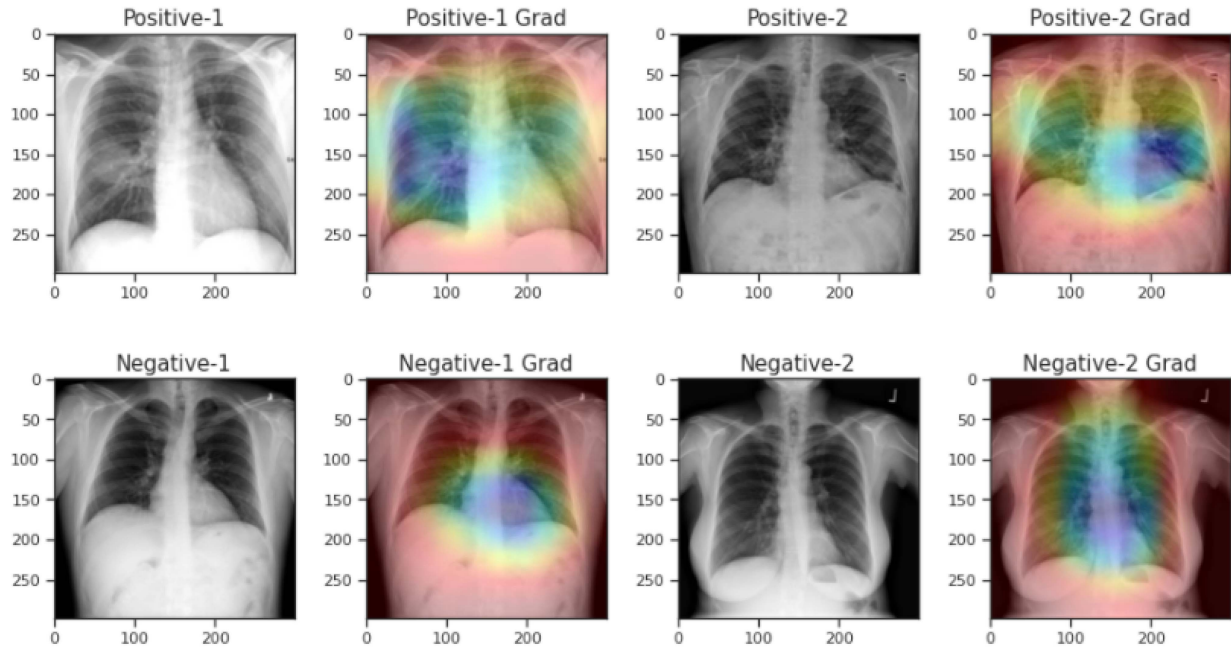


Fig. 4. Classification and Grad-CAMP results, the first row shows the results for positive samples, while the second row shows the results for negative samples.

be obtained from 64 filters of size $64 \times 3 \times 3$. The total trainable parameters are 315,058.

C. Explaining Deep Learning Features Using Gradient-Weighted Class Activation Mapping (Grad-CAM)

To understand and visualize the essential characteristics of classified medical images, Gradient-weighted Class Activation Mapping (Grad-CAM) is used, as shown in Fig. 3. It employs gradient information and provides an explainable view using the final convolutional layer results to construct rough localization maps and emphasize the significant regions in the input image to indicate the details. The method is an advancement over prior techniques in versatility and precision.

From a high level, an image as input is taken and created as a model that is trimmed off at the convolutional layer for which we want to construct a Grad-CAM heat map. Next, we connect the fully-connected layers for classification and prediction. We then manipulate the input through the deep learning architecture, maintaining the layer output and loss. Following, we locate the gradient of the output features of the desired layer concerning the model loss. The gradient contributes to the prediction, reduction, resizing, and rescaling sectioned so that the heat map can be overlapped with the initial image. The Grad-CAM carries a simple input image, and detection methods are used further by utilizing the developed model. Once the expected labels have been estimated utilizing the complete architecture, Grad-CAM is used for any of the selected convolutional layers. Mainly, the final layer is utilized for applying Grad-CAM.

IV. RESULTS

The visualization results of Grad-CAM results are shown in Fig. 4. The first row shows the results for Positive samples, the Positive-1 Sample Insight: In this Figure, Grad-CAM image on the right mid part of it we can see the blue colour highlighted

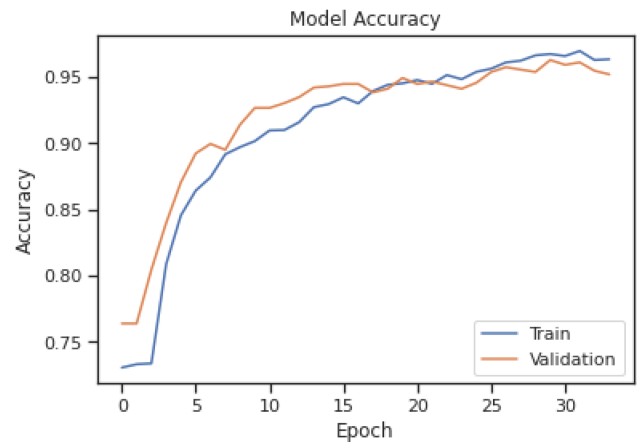


Fig. 5. Training and Validation accuracy of the proposed model, it can be seen that accuracy is significantly improved after 5th epoch.

portion, which is opacity due to which it belongs COVID - Positive Category. In Positive-2 Sample Insight: its Grad-CAM image on the left bottom part of it, we can see the blue, green colour highlighted portion, which is the consolidation and no Tree-Bud due to which it belongs to COVID the Positive Category. Negative-1 Sample Insight: In its Grad-CAM image, we can see the blue colour highlighted portion, which is between the Cardiac and Diaphragm, and no opacity is detected, due to which it belongs to COVID - Negative Category. Negative-2 Sample Insight: In its Grad-CAM image, we can see the blue colour portion, which highlights the Trachea, and no other opacity is detected due to which it belongs to COVID - Negative Category. The Precision, Recall and F1-score results per class of the above proposed model are shown in Figs. 8, 9, and 10, respectively. In Fig. 8, the Precision results are 69.7% for COVID-19 images, while for normal class, it is 88.7%. The Precision results for both types are 79.2%. The Recall result is 89.5% for standard images,

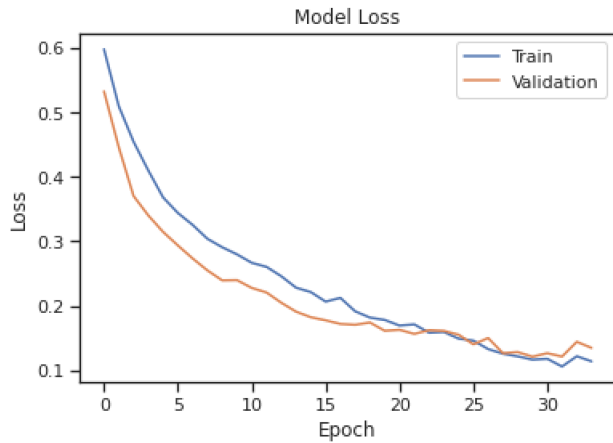


Fig. 6. Training and Validation loss values, after the 20th epoch, the value decreases, which shows the efficiency of the proposed model.

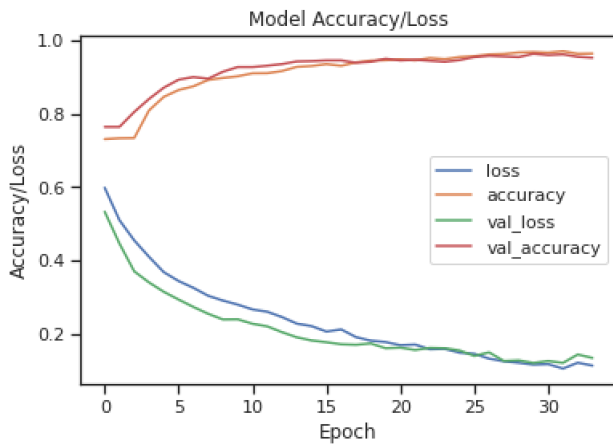


Fig. 7. Training & Validation accuracy's/loss curve shows improvement in model's performance over time.

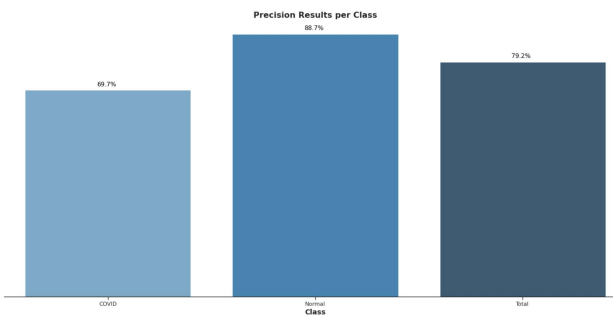


Fig. 8. Precision results per class.

which is high for normal images as compared to COVID-19 images. The combined results are 78.7%. The F1 results of normal images are 89.1% which is relatively high compared to Precision and Recall. It can be observed that results are high for the normal class. The overall results for both types are shown at a third bar; it can be seen that the results are good and nearly equal to 80%.

The experiments are conducted on an Intel Core-i5 desktop with 24 GB RAM, a 3.2 GHz CPU, and an NVidia GeForce GTX 1080 Ti Graphics Processing Unit (GPU). The programming



Fig. 9. Recall results per class.

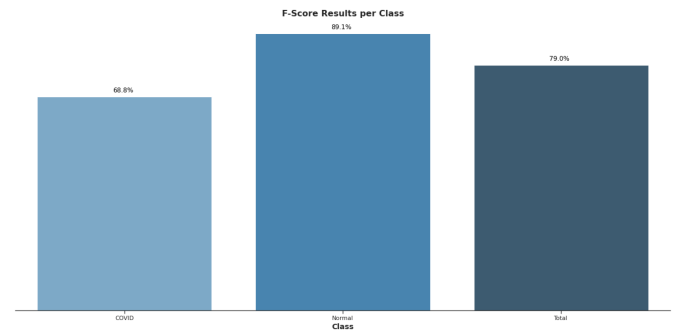


Fig. 10. F1 Score results per class.

language is Python with different libraries, including Keras, Numpy, Tensorflow, and others for visualization and performance monitoring, like Scikit-Learn, OpenCV, etc. The programming conditions are configured on Windows 10 operating system. The training process is set for 40 epochs with a batch size of 128 and a learning rate of 0.01. Adam optimizer is employed to handle noisy data. The validation data is used during training to check the model's learning gain. The data sets utilized in the experiments are collected from online repositories [13]. The data set comprises 6,334 chest scans for training, which were de-identified to protect patient's privacy. We have used 4243 images for training and 2091 images for testing purposes. All images are in DICOM format and are labeled for the presence of opacities and overall appearance by a panel of experienced radiologists.

Fig. 5 demonstrates the progress of the training and validation process, and it is evident that the model delivers smooth generalization on the data set while achieving almost 98% Training-Validation accuracy. The accuracy value is improving with more training. Fig. 6 exhibits 0.15% Training-Validation loss, which means that the model is improving its performance with more learning or training. We observe that Train and Validation Accuracy's Curve slightly overlaps and is the same as Loss Curve as shown in Fig. 7. Hence, Overfitting is avoided; this is possible because Dropout Regularization and Early Stopping Metrics are applied in this work. Moreover, looking at the accuracy and loss curve, the model performance is growing over time, which means the model is improving with training.

Furthermore, we also performed a comparison with the state-of-the-art deep learning-based model. The results are shown in Table I. From the data shown in Table I it can be seen that the

TABLE I
COMPARISON RESULTS WITH STATE-OF-THE-ART CNN MODELS

S.NO	Method	Accuracy	AUC
1	VGG-16	0.93	0.90
2	ResNet-50	0.94	0.93
3	Inception	0.94	0.94
4	CapsNet	0.95	0.94
5	Ours CNN	0.96	0.96

overall performance of the model is good, with an accuracy rate of 0.96, as compared to pre-built CNN models, namely, VGG-16, ResNet-50, Inception, and CapsNet. These models achieve accuracy rates of 0.93, 0.94, 0.94, and 0.95 for the classification of COVID-19 chest X-ray images. To show the performance of the proposed model we also compared the Area Under the Curve (AUC) of the proposed model with others. It can be seen the value of the proposed model is 0.96, which is good and shows the capability of models in distinguishing different classes of Chest X-ray images.

V. CONCLUSION

This paper offered an intelligent blockchain and an AI-enabled solution for the healthcare industry. The developed system aids in controlling the coronavirus pandemic. To further incorporate Blockchain technology, a new deep learning-based architecture is designed to identify the virus in medical images. The blockchain-enabled the developed system to offer reliable data-gathering platforms and promising security solutions, thus assuring the high quality of COVID-19 data analytics. We proposed a sequential deep learning architecture with numerous layers using a benchmark data set. With a classification accuracy rate of 0.96, the architecture produces good results. This research may be extended in the future to analyze and segment other viral infections; we may utilize different segmentation approaches based on deep learning applications for the analysis, segmentation, detection, and classification of various viral diseases.

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